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Previews

Minding the gaps: The importance of navigating holes in protein fitness landscapes

Neil Thomas[1] and Lucy J. Colwell[2,3,*]
[1]Computer Science Division, University of California, Berkeley, Berkeley, CA 94720, USA
[2]Department of Chemistry, University of Cambridge, Cambridge CB2 1EW, UK
[3]Google Research, 1600 Amphitheatre Parkway, Mountain View, CA 94043, USA
*Correspondence: ljc37@cam.ac.uk
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Machine-learning-guided protein design is rapidly emerging as a strategy to find high-fitness multi-mutant variants. In this issue of *Cell Systems*, Wittman et al. analyze the impact of design decisions for machine-learning-assisted directed evolution (MLDE) on its ability to navigate a fitness landscape and reliably find global optima.

As Maynard-Smith pointed out in 1970, evolution has generated functional proteins largely through small steps—by making one or perhaps two changes at a time to the sequence of an existing protein or protein domain and keeping those changes that do not impair function (Smith, 1970). Similarly, directed evolution has excelled at identifying sequence variants with improved functional activity, typically by following paths in sequence space along which protein function is maintained (Arnold, 2019). However, protein fitness landscapes are known to be rugged, containing many local optima that can trap greedy strategies such as directed evolution and prevent them from finding better optima (Poelwijk et al., 2019).

Model-guided protein design presents a potential solution to this challenge, whereby a model is built that captures the fitness landscape, i.e., the relationship between sequence and the functional activity of interest. This model is then used to guide the experimental exploration of sequence space (Romero et al., 2013). A model that accurately captures the relationship between sequence and measured activity can be used to escape local optima and teleport across valleys in the fitness landscapes to distant peaks. Such models thereby open up vast regions of sequence space that were previously inaccessible because they could not be reached via paths that maintain function.

While there are many ways in which such models might be built, recent strides have been made using machine learning (ML) to fit models to limited experimentally labeled data (Romero et al., 2013). Model performance is evaluated using sequences that were held out from model training, which can be chosen from the labeled data at random or in such a way as to test the ability of the model to extrapolate, e.g., by choosing those labeled sequences with highest measured fitness values or that are furthest from the wild-type starting sequence as test data (Bryant et al., 2021). The best performing model(s) are then used to design new sequence variants, which are synthesized and tested. In a multi-round setting, an active learning strategy would take these data points and use them to update and improve the ML model through additional design-build-test iterations.

Experimental results suggest that despite the lack of mechanistic or physics-based knowledge explicitly included in such models, they are nonetheless able to navigate sequence space and identify novel functional sequence variants in which more than 50% of the residues are mutated from the starting sequence. However, numerous design decisions are required, such as the choice of model architecture and the size of the initial dataset (Sinai & Kelsic, 2020). For many of these decisions, the choice that yields the best outcome will vary depending on details of the biological system of interest. Identifying strategies that generalize across different systems is very much an open challenge in the field (Angermueller et al., 2020).

In this issue of *Cell Systems*, Wittman et al. (2021) identify an exciting example of a generalizable strategy. Working with an approach termed machine-learning-assisted directed evolution (MLDE) (Wu et al., 2019) in the context of a dataset that contains nearly all amino acid combinations at four positions (Wu et al., 2016), Wittman et al. observed that the experimental fitness landscape is dominated by variants with extremely low or zero fitness, which they term "holes." This phenomenon is not unique to this dataset: in general, the prevalence of such holes is exacerbated by the fact that experimental fitness assays often have a lower bound or floor on the fitness level that they can measure, such that they provide no quantitative information about how bad a variant is below this floor.

In the dataset analyzed by Wittman et al., the experimental label for each variant is determined via the ratio of sequence counts after selection to sequence counts in the input library (Wu et al., 2016). Of the 149,361 labeled variants, 29,477 were reported to have zero counts after selection, while 87,073 variants have 5 or fewer sequence counts after selection—with small numbers, Poisson noise can significantly affect the reported fitness. As Wittman et al. explain, including these data points in a fixed-size training set reduces the amount of information available to train a regression model. For example, consider two variants A and B that are both labeled with the minimum fitness measurement, where variant B includes an additional mutation compared to the sequence of variant A. The measurements only reveal that the additional mutation is not enough to rescue the variant and push it above the

fitness floor—they cannot reveal whether B is slightly better or slightly worse than A, in contrast to the case in which both variants are assigned quantitative labels.

Wittman et al. exploit the near-complete nature of this dataset to empirically measure how the presence of holes in the training data affects the performance of the resulting trained model. Fixing the size of the training set at 384 data points, they demonstrate that varying the composition to reduce the number of data points with extremely low or zero fitness significantly improves the performance of MLDE, as reflected by a number of different metrics computed over data from 2,000 simulations of the MLDE protocol. For example, requiring that 50% of the 384 training data points have fitness above a comparatively low threshold (just over 1% of the global maximum) led to a 49% improvement in the mean fitness achieved by the top 96 predictions made by the resulting model. Wittman et al. also explore different data encodings and model architectures and report much smaller effects on model performance.

The next key observation made by Wittman et al. is that zero-shot modeling approaches can be used prospectively to reduce the presence of holes in the training data. Zero-shot modeling approaches take advantage of physics-based models or ML models fit using a large corpora of protein sequences to predict the functional activity of sequence variants without requiring access to any experimental measurements. Wittman et al. demonstrate that a number of these approaches can be used to pre-filter, or focus, the training data and thereby greatly reduce the presence of holes. They term this strategy "ftMLDE" and show that it greatly enhances performance, enabling the global maximum to be attained in 99.7% of simulations; in comparison, traditional DE attains the global maximum in just 1.23% of simulations, while MLDE without pre-filtering attains this maximum in 8.85% of simulations.

Application of this innovative strategy to future wet-lab engineering campaigns is greatly facilitated by the comprehensive and highly accessible software package provided by Wittman et al. to accompany their work. The significant improvement attained by this elegant strategy, together with its ease of use, will enable researchers to leverage this approach for a wide range of protein engineering and design challenges.

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DECLARATION OF INTERESTS

N.T. and L.J.C. performed research while at Google LLC.

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